

Conjecture 00000003794: An LCP with infinitely many solutions

C0ldSmile

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Abstract

A nonzero positive semidefinite matrix of order two has an entire ray of linear complementarity solutions. This disproves the unrestricted solution-count bound 2^n stated in conjecture 00000003794. Lean verifies the standard LCP conditions, the complete solution set, its infinite cardinality, and the negation of the universal bound.

1 Source statement and scope

The English conjecture reads:

Definition: Linear complementarity problems: complementarity structures on cones.
Conjecture: The solution count of LCP: the upper bound of solution counts of linear complementarity is 2 to the n-th power, with the tightness of bounds the strictness of P-matrices. (LCP P matrix strict solution upper bound tightness)

The exact bilingual source is included as `conjecture.md`. In both languages the first clause states a solution-count bound without assuming a matrix class, nonsingularity, or finitely many or isolated solutions. The subsequent P-matrix tightness phrase supplies no precise additional definition. We refute the explicit unrestricted first clause. Our matrix is singular and is not a P-matrix; this is not a disproof of a reformulation restricted to P-matrices.

2 The complete counterexample

For a real $n \times n$ matrix M and $q \in \mathbb{R}^n$, the standard orthant LCP has solution set [1, p. 540]

$$\text{SOL}(M, q) = \{x \in \mathbb{R}^n : x \geq 0, \quad Mx + q \geq 0, \quad x^\top(Mx + q) = 0\},$$

where inequalities are coordinatewise. The nonnegative orthant is a cone, so this is already a case of complementarity on cones.

Theorem 1. *Let $M = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $q = 0$. Then $\text{SOL}(M, 0) = \{(t, t) : t \geq 0\}$, an infinite set. In particular, it has more than $2^2 = 4$ solutions, although M is nonzero and positive semidefinite.*

Proof. For $x = (a, b)$, $Mx = (a - b, b - a)$. Feasibility requires $a, b \geq 0$, $a - b \geq 0$, and $b - a \geq 0$. Thus $a = b = t \geq 0$. Conversely, every (t, t) with $t \geq 0$ has $Mx = 0$, so all three LCP conditions hold. The map $k \mapsto (k, k)$ from natural numbers into this solution set is injective. Already $(0, 0), (1, 1), (2, 2), (3, 3), (4, 4)$ are five distinct solutions. Finally $M \neq 0$, M is symmetric, and $x^\top Mx = (a - b)^2 \geq 0$. Its determinant is zero. $\square$

3 Solutions and complementarity patterns

The 2^n choices of a zero coordinate from each pair $(x_i, (Mx + q)_i)$ need not determine distinct, unique vectors. Here $Mx = 0$ leaves an entire ray. Rohn’s Proposition 2.4 gives a 2^n solution bound assuming every complementary matrix is nonsingular; Proposition 2.5 describes infinite families [2]. These results provide context, not premises for our elementary counterexample.

Since $\det M = 0$, this matrix fails the P-matrix condition that all principal minors are positive. We count all solutions; we make no claim about bounds restricted to isolated solutions or to finite solution sets. For $t > 0$, each $x_i = t > 0$ and each residual coordinate is zero, so these points are even strictly complementary. Strict complementarity of a solution is different from matrix nonsingularity.

4 Formal statement correspondence

In `Problem.lean`, the orthant is Mathlib’s `ConvexCone.positive`. The definition `LCPSolutions` uses actual matrix-vector multiplication, nonnegativity of both vectors, and their zero dot product. The coordinatewise equivalence theorem proves the standard scalar LCP conditions: a finite sum of nonnegative products is zero exactly when every product is zero.

The matrix and diagonal vectors are concrete real matrices and functions on `Fin 2`. The theorem `counterSolutions_iff` classifies all solutions. Lean also verifies nonzeroness, determinant zero, symmetry as the real Hermitian condition, the quadratic identity, and Mathlib’s `Matrix.PosSemidef` predicate.

The root module proves injectivity of the natural-number family and infinitude of the actual solution set. Its `Set.encard` is the extended natural cardinality, which equals infinity here. Bare natural-valued cardinality would assign zero to an infinite set and misstate the bound. The predicate `SolutionCountBound` is

$$\forall n \in \mathbb{N}, n > 0 \implies \forall M \in \mathbb{R}^{n \times n} \forall q \in \mathbb{R}^n, \text{ encard}(\text{SOL}(M, q)) \leq 2^n.$$

The theorem `conjecture3794_disproof` negates this predicate at $n = 2$, the displayed matrix, and $q = 0$. All connections to the actual solution set are proved.

5 Reproduction and verification

Lean 4.19.0 and Mathlib v4.19.0 are pinned with exact dependency revisions. The README lists commands for `lake build` and strict replay of all three Lean sources with warnings treated as errors. `Check.lean` prints six definitions and audits all 23 theorem types and axiom dependencies.

This package records a fresh independent build, strict replay, internal semantic review, and rendered-PDF inspection. No numerical auxiliary program is needed. These local checks are separate from maintainer acceptance. Source and rules were checked at revision 0862407e; eligibility evidence records unsolved metadata and prior-submission searches.

References

- [1] X. Chen and S. Xiang. *Sparse solutions of linear complementarity problems*. Mathematical Programming 159 (2016), 539–556. Author-hosted article, §1.
- [2] J. Rohn. *Description of all solutions of a linear complementarity problem*. Electronic Journal of Linear Algebra 18 (2009), 246–252. Journal article, Propositions 2.4–2.5.